HARSHAVARDHAN REDDY NARRA

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MS Electrical Engineering student at USC specializing in VLSI and Computer Architecture, with research experience in CNN accelerators, HDC hardware pipelines, and full-custom layout. Skilled in RTL design, FPGA implementation, and functional verification across academic and research environments.

EDUCATION

University of Southern California August 2025 – Present

Master of Science in Electrical Engineering (VLSI and Computer Architecture)

- Relevant Coursework: EE457 – Computer Systems Organization, EE477 – MOS VLSI Circuit Design

Jawaharlal Nehru Technological University Hyderabad August 2021 – May 2025

Bachelor of Technology in Electrical and Electronics Engineering

- Relevant Coursework: Computer Organization, Digital Electronics, Computer Arithmetic, Microprocessor and Microcontroller

PUBLICATIONS

- Presented paper at ROCS 2024, HiPC (IEEE International Conference on High Performance Computing, Data, and Analytics): "Efficient Feature Extraction for ViT Model using Custom CNN Accelerator." DOI: 10.1109/HiPCW63042.2024.00016
- Participated in DVCON India 2024 Conference – Architecture to Analytics (A2A) track

TECHNICAL SKILLS

Architecture / Modeling: Computer Architecture, Microarchitecture Design, Performance Modeling

RTL / EDA: Verilog, SystemVerilog, RTL Design, Cadence Virtuoso, QuestaSim, Vivado

Software: C++, Python, TCL, Linux

EXPERIENCE

Indian Institute of Technology (IIT) Bhubaneswar, Odisha, India January 2026 – Present

Research Intern

- Designing a synthesizable SystemVerilog RTL pipeline for Hyperdimensional Computing (HDC), implementing a 1024-bit XOR-plus-permute datapath with configurable permutation modes (word reversal, per-word rotation, full-vector rotation)
- Built a parameterized, modular architecture (WORDS x BITS_PER_WORD) with valid/ready handshake signaling and registered pipeline stages for clean timing closure
- Developed a self-checking SystemVerilog testbench with golden-model verification covering all permutation modes, boundary rotation values, and output backpressure stall scenarios

Indian Institute of Technology (IIT) BHU, Uttar Pradesh, India May 2023 – July 2023

Summer Research Intern

- Developed an IoT-based wearable health monitoring device on ESP32-WROOM with integrated PPG, ECG, temperature, and gas sensors; implemented real-time signal processing, wireless transmission, and optimized power management with a team of 6
- Improved diagnostic accuracy by 8% for aerospace medicine applications

ACADEMIC PROJECTS

5×6 Systolic Array-based CNN Accelerator using Verilog

- Devised a LeNet-5-inspired CNN accelerator IP on Digilent Genesys 2 FPGA (Kintex-7, XC7K325T-2FFG900C)
- Implemented convolution, pooling, and activation modules in Verilog, integrated with Vega AS1061 RISC-V processor
- Achieved 116 GOP/s throughput at 200 MHz with 2.498 W power; optimized FPGA utilization (65% LUTs, 18% BRAM, 11.5% DSP)

Efficient Feature Extraction for ViT-based Malware Detection

- Leveraged custom CNN accelerator to accelerate ViT-based malware detection workloads with a team of 3
- Reduced feature-extraction latency by 2.8× compared to baseline while ensuring ViT prediction accuracy remained unchanged after integration
- Earned First Runner-Up distinction at the DVCON India Design Contest

5-Stage Pipelined RV64I MIPS Processor

- Designed a 5-stage RV64I processor (SystemVerilog) with Zba support, forwarding, ID-bypass, load-use stall, and branch control logic
- Built a self-checking testbench with assertions to verify ALU ops, lw/sw, beq, and Zba functionality under hazard conditions
- Debugged and validated pipeline timing and RAW hazard handling using ModelSim waveform analysis

Full-Custom 16-bit MAC Design | Cadence Virtuoso

- Designed a full-custom 16-bit MAC unit (Booth + compressor tree + sparse-4 Kogge-Stone) and completed end-to-end schematic-to-layout implementation with DRC/LVS and post-layout verification
- Achieved 1.2 ns clock period (~833 MHz) with 577.3 µW power consumption and 2041 λ^2 layout area; optimized routing congestion and PPA through architectural trade-off analysis